



RAJESHKUMAR K

Associate DevOps Engineer · Observability (Logging & Monitoring)

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PROFESSIONAL SUMMARY

DevOps engineer with 1.5+ years in production support and observability at Trinity Mobility, a smart-city and IoT platform company. In the Technical Assistance Center (TAC) I own logging and monitoring (LnM) for 20–25 live client projects and fix capacity and health problems at the warning stage, before they become production incidents that hit the company SLO. I am currently leading the R&D effort to bring logs, metrics and traces for the whole platform into one place with **OpenObserve** and **OpenTelemetry**, from Kubernetes infrastructure and Java microservices to data-lake engines and Gen AI workloads.

20–25 production projects monitored	14 DLH service dashboards built	83 real-time alerts on 47 health metrics	70 / 80 / 90% warning / critical / disaster thresholds
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TECHNICAL SKILLS

Observability	OpenObserve, OpenTelemetry Collector (agent + gateway), Grafana, Prometheus, VictoriaMetrics, InfluxDB, ELK Stack (Filebeat, Logstash, Elasticsearch, Kibana), Alertmanager, Uptime Kuma, Langfuse
Tracing & telemetry	Distributed tracing, trace-log correlation (trace_id in logs), OTLP logs, Spring Boot Actuator, JMX, cAdvisor, kube-state-metrics, node / Postgres / MSSQL / MQTT / Kafka exporters
Containers & delivery	Kubernetes (on-prem, Azure AKS), Helm, ArgoCD (GitOps), Docker, Nginx
Platforms monitored	Kafka, Apache NiFi, WSO2 ESB / API Manager, Trino, Spark, Livy, Airflow, Druid, MinIO, Ranger, OpenMetadata, Hasura, MobilityDB, LDAP, Redis, Vault
Gen AI observability	NVIDIA GPU monitoring (nvidia_gpu_exporter), LiveKit voice AI / SIP call quality, Langfuse, LiteLLM, n8n agent workflows (model, token and cost tracking)
Scripting & query	Bash, PowerShell, Python, SQL, PromQL
Operations	Jira, severity-based alerting, SLO protection, root-cause analysis, runbooks, Linux, Git

PROFESSIONAL EXPERIENCE

Trinity Mobility Pvt Ltd — Associate DevOps Engineer Mar 2025 – Present
Technical Assistance Center (TAC): post-go-live DevOps support and Logging & Monitoring for the company's smart-city and IoT projects

Production support and LnM ownership

- Own logging and monitoring for **20–25 go-live projects**. Every production alert from the LnM stack raises a Jira ticket: I fix the ones in my area, and assign, follow up and close the ones that belong to other teams.
- Run severity-based alerting (**Warning 70%, Critical 80%, Disaster 90%+**) so that disk, memory, CPU and service-health problems are fixed before they turn into production events that affect the company SLO.
- Run and support the **LnM monitoring services** that every project depends on: Prometheus, VictoriaMetrics, Grafana, ELK, InfluxDB, Alertmanager (email and Telegram), Uptime Kuma, OpenTelemetry collectors, kube-state-metrics and node / Postgres / MSSQL / MQTT / Graphite exporters.
- Run the centralised **ELK logging platform** on Kubernetes, including index management, Logstash security and storage tuning for reliable log retention.
- Root-caused and fixed production issues, including a Kafka exporter that could not serve metrics for 4,900+ topics and 44k partitions, an SSL (PKIX) failure after a partner changed certificate authority, health checks breaking on non-numeric statuses, and metric scraping lost after GitOps syncs removed pod annotations.

OpenObserve R&D: unified observability platform (end to end)

- Installed and built a complete **OpenObserve** stack on Kubernetes, with OpenTelemetry agent and gateway collectors (push and pull split), Kubernetes events, node and container metrics, and OTLP logs, so that every issue can be investigated in one place.
- Enabled **distributed tracing with trace-log correlation** for Java microservices (log4j MDC), WSO2 ESB and API Manager, NiFi, CEP, LDAP, IDAM and data-lake services. End-to-end trace checks passed 50/50 on the nightly environment and 14/14 on R&D.
- Built **trace-level dashboards in OpenObserve**: request rate, errors and latency per service and operation from span metrics, slowest calls, and drill-down from a panel to the trace and its correlated logs.
- Designed **45 alert pipelines and 83 real-time alerts** over 47 resource-health metrics, grouped into 9 categories and routed to Telegram, email and incident management.
- Kept the OpenObserve platform stable: diagnosed scheduler, memory and storage failures, and wrote runbooks for the team.

Data Lake House (DLH) service dashboards in Grafana (key achievement)

- Built the **OSS Engine Overview** and **13 per-service DLH dashboards** (Airflow, Trino, Spark, Livy, Kafka, Druid, MinIO, Ranger, OpenMetadata, Hasura, InfluxDB, MobilityDB and more). Each one shows UP / DOWN / UNREACHABLE health, latency, JVM and pod health on a single screen.
- Added the missing collector scrape jobs and exporter fixes that feed these dashboards, and migrated 13 OpenObserve dashboards to Grafana.

1999 vendor API: full end-to-end trace (key achievement)

- Traced the **1999 emergency-incident API** as one trace across the API gateway, WSO2 ESB, Vault, the IoT operations service and Redis, and the command-centre application, including the background fire-station recommendation flow.
- Used per-hop timings to find delays and to confirm the impact of sending the caller's reply earlier in the flow.

Gen AI observability

- Set up **Langfuse** (self-hosted, behind SSO) for LLM observability across the agentic AI, voice AI and guardrails projects, tracking prompts, responses, latency, token usage and cost.
- Instrumented **n8n** AI-agent workflows with OpenTelemetry. The collector sends each trace to both Langfuse and OpenObserve, with model, tokens, cost, workflow, agent and node details.
- Connected the **LiteLLM** model gateway to Langfuse, so every model request is tracked by provider, model and error type.
- Built the **LiveKit voice AI production dashboard** (28 panels): active rooms, participants and SIP calls; call quality (media quality score, call-setup latency P95, jitter P95, packet loss, RTT P95, WebRTC connection failures, retransmissions); SIP trunk status, error rate, terminations by reason and transfers; track publish/subscribe success rates.
- Built **NVIDIA GPU monitoring** for the GenAI servers (Windows and Linux) with `nvidia_gpu_exporter` and Prometheus: GPU and memory utilisation, memory allocation, temperature, power draw, SM/graphics/memory clock speeds, P-state and throttle reasons, alongside CPU, memory, disk and network health.
- Built an **n8n system health** dashboard: active workflows, event-loop latency, heap and GC, memory, uptime and leader status.

Other contributions

- Built a port-monitoring dashboard, Airflow DAG/task dashboards and microservice health dashboards (JVM, Python, HikariCP), plus an IoT packet-journey analysis across Kafka, NiFi and CEP.
- Wrote 20+ study guides, playbooks and runbooks (OpenObserve alerting, monitoring setup, tracing, certificate fixes) that the team uses for training and incident handling.

EDUCATION

B.E. Mechanical Engineering — Kongu Engineering College, 2019
CGPA 9.01 / 10

Languages: English (fluent), Tamil (native), Hindi (basic)

CERTIFICATIONS

- Microsoft Certified: **Azure Fundamentals (AZ-900)**
- DevOps Engineer Internship — IBM with StarAgile, 2024
- NCC "C" Certificate — Cadet Under Officer, 2019